

IYS



CLOSED-LOOP NEMA 17



SHIPS IN 2-3 BUSINESS DAYS



BOXPOX

BUY

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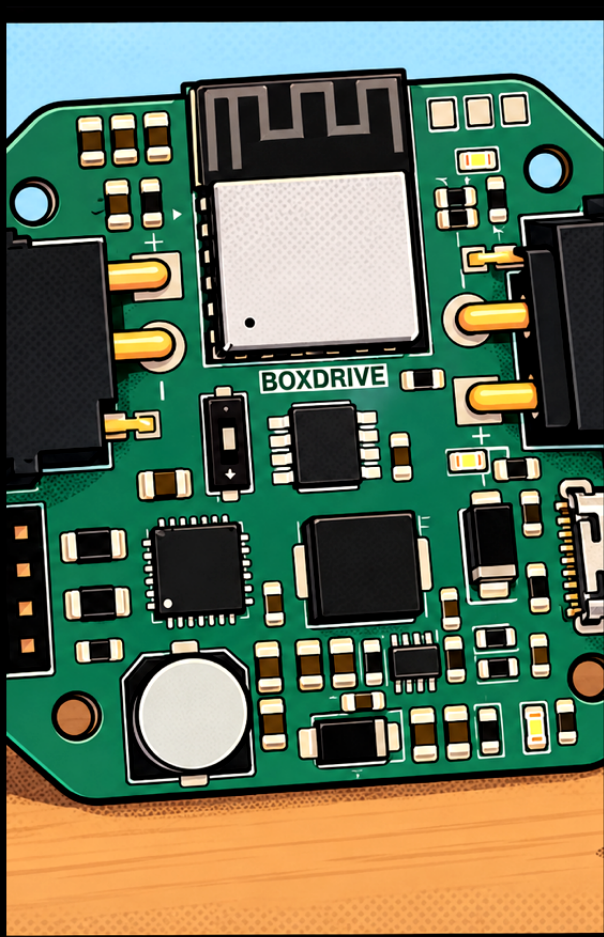
BOXDRIVE



Closed-loop stepper controller for
makers, roboticists, and many more
use cases

BUY

FEELS LIKE HOME. ✨



boxdrive.ino

```
#include <BOXDRIVE.h>

BOXDRIVE motor;

void setup() {
  motor.begin();
  motor.setCurrent(800);
}

void loop() {
  motor.moveTo(90);
}
```

```
Everything working fine! ·
BOXDRIVE on COM3
```

ROBOTICS

CNC

WIRELESS

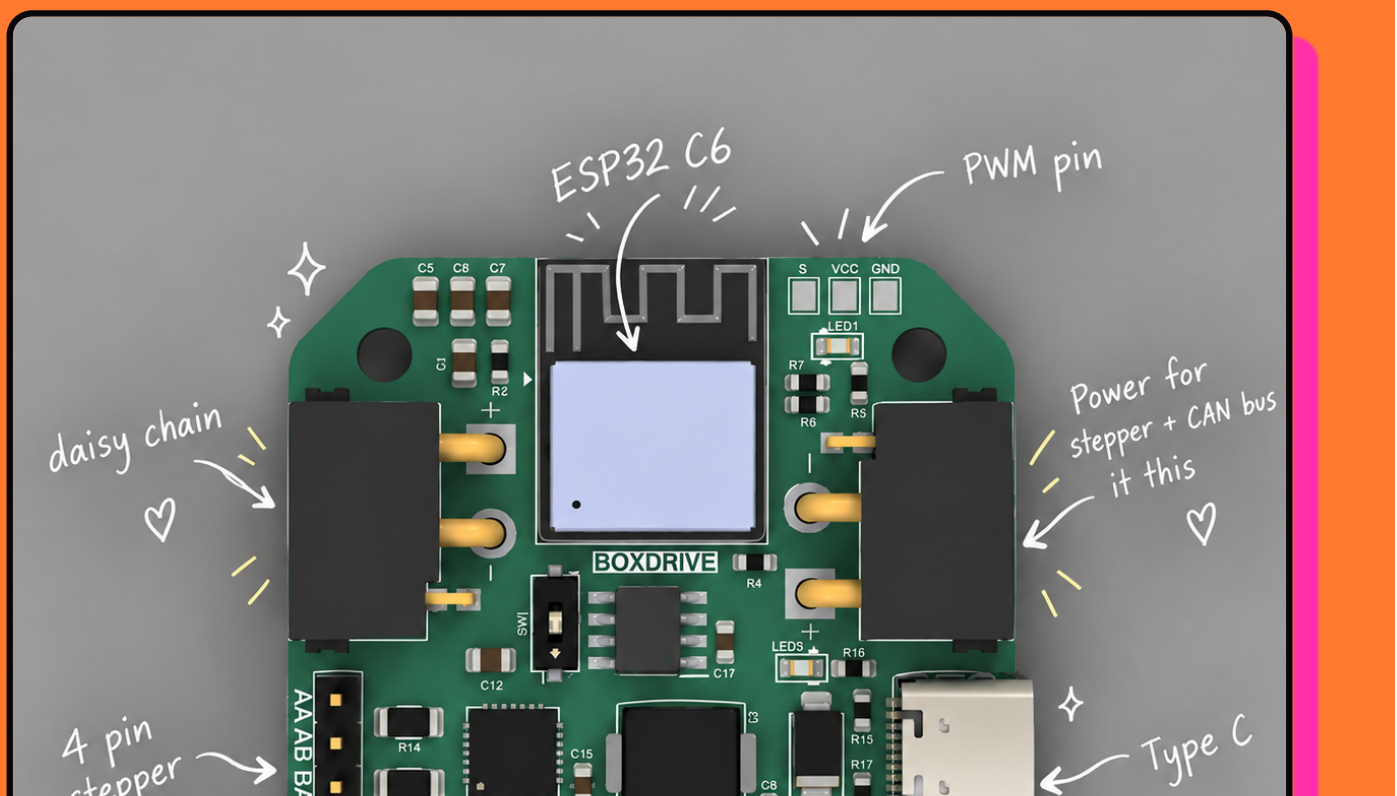
JOINTS THAT NEVER LOSE A STEP.

Closed-loop control on every NEMA 17. The AS5600 watches the shaft, the TMC2209 whispers, and the ESP32-C6 keeps the whole arm in sync over CAN.

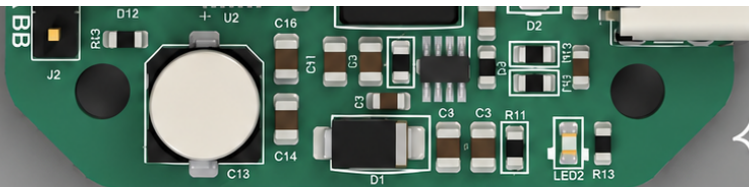
FROM IDEA TO MOTION IN MINUTES.

Don't spend a weekend wiring a driver, encoder and MCU on a breadboard. Bolt BOXDRIVE to the back of a NEMA 17 and start writing firmware.

Start right over



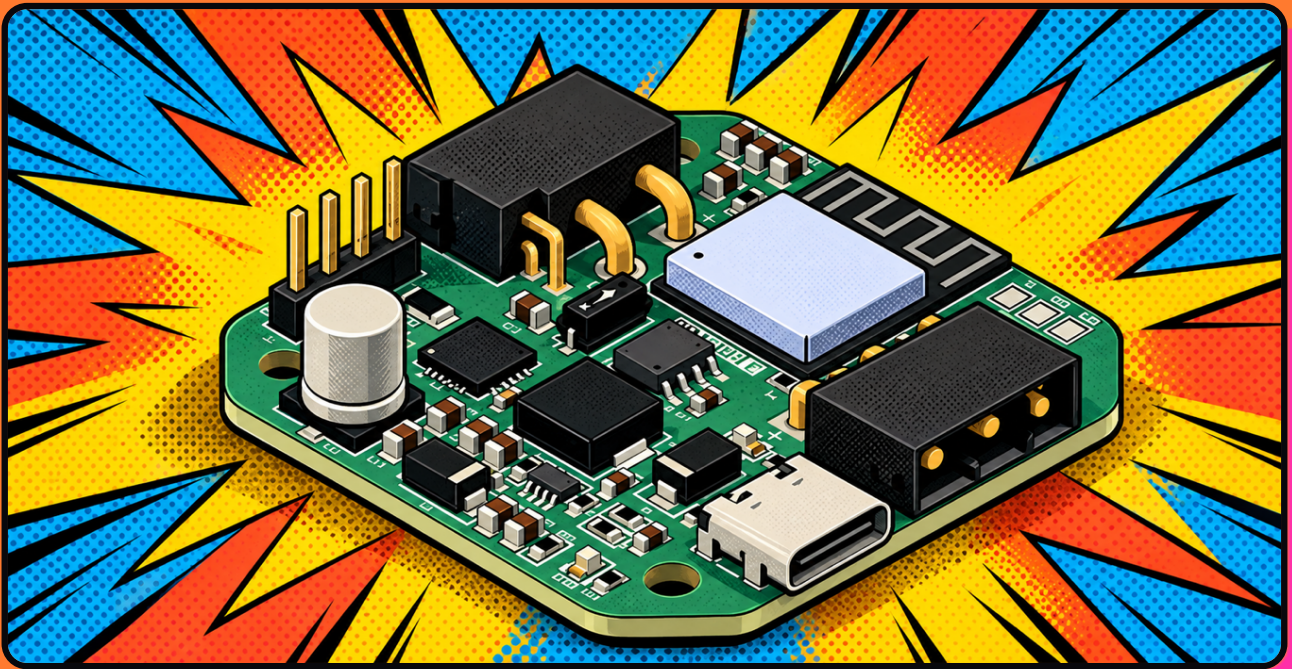
connector



BoxDrive

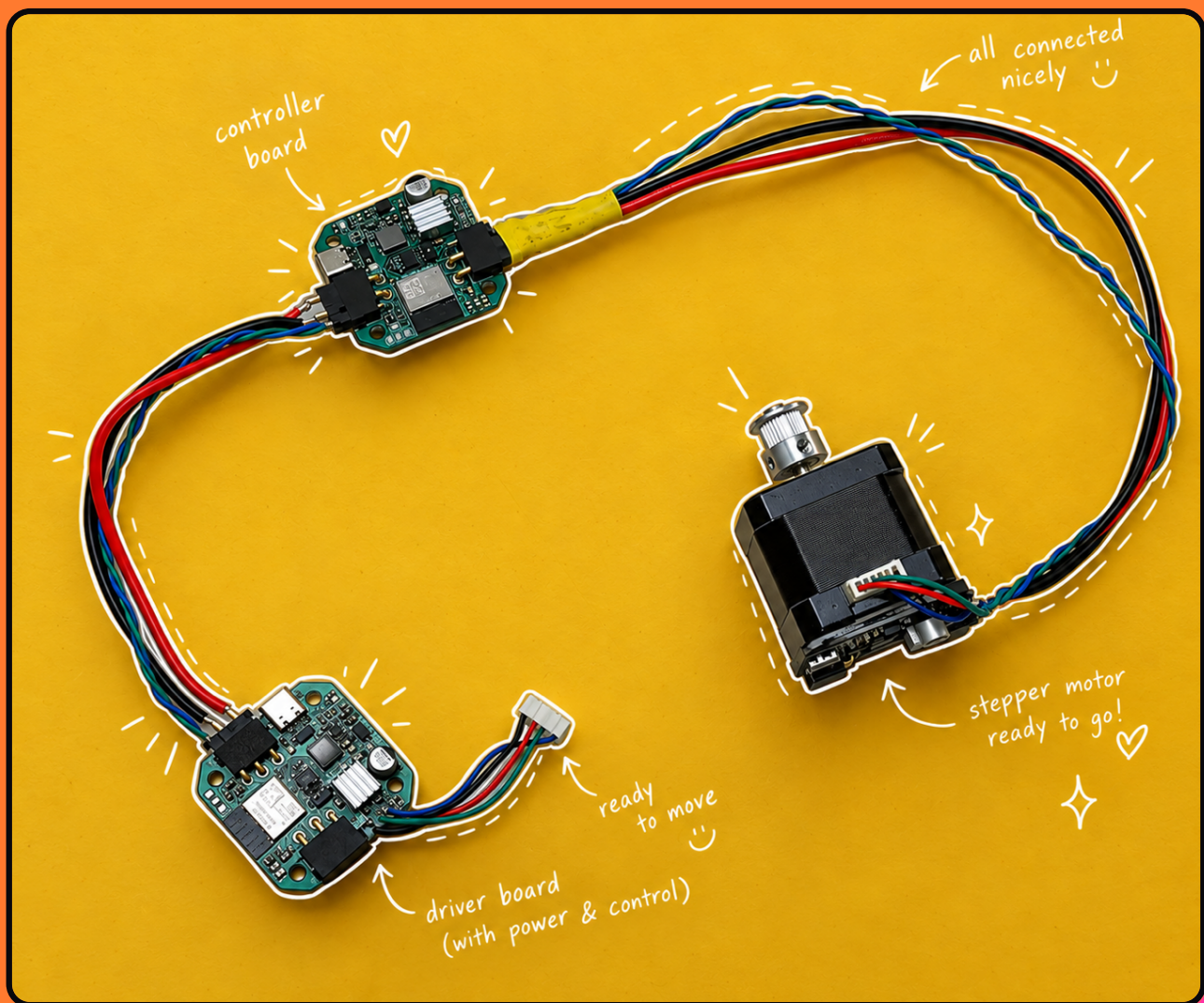
KNOW EXACTLY WHAT YOU'RE DOING.

With clear silkscreen indicators and logically grouped I/O, BOXDRIVE makes it immediately obvious where everything goes. The onboard ESP32-C6 gives you the freedom to build exactly what you want.



DAISY CHAIN FOREVER.

Building a multi-axis machine? Connect motors back-to-back using the CAN bus and the high-current XT30 connectors to completely eliminate wiring spaghetti. One cable for power, one for data.



SILENCE IS A SPEC.

StealthChop

SpreadCycle

2.0 A RMS

TMC2209 keeps the motor quiet even under load. UART configurability for current, microsteps and StallGuard.

ABSOLUTE POSITION.

AS5600

12-bit

0.0879°

A magnet on the shaft, an encoder on the back of the board. Closed-loop without an extra sensor harness.

WIRELESS FROM DAY ONE.

Wi-Fi 6

BLE 5

CAN

ESP32-C6 RISC-V at 160 MHz. Program over USB-C like any ESP32, then update the fleet over the air.

SMALL BOARD. BIG TOOLBOX.

Every chip, connector and rail BOXDRIVE packs in.

ESP32-C6 RISC-V

160 MHz wireless MCU with 4 MB flash and 512 KB SRAM.

TMC2209 DRIVER

Silent StealthChop / SpreadCycle, UART setup, up to 2.0 A RMS.

AS5600 ENCODER

12-bit magnetic rotary sensing at 0.0879° resolution.

4.75-24 V INPUT

XT30 daisy-chain power for multi-axis machines.

CAN BUS

SN65HVD230 transceiver so axes talk without a wiring nest.

WI-FI 6 + BLE 5

On-chip radio with PCB antenna. OTA capable.

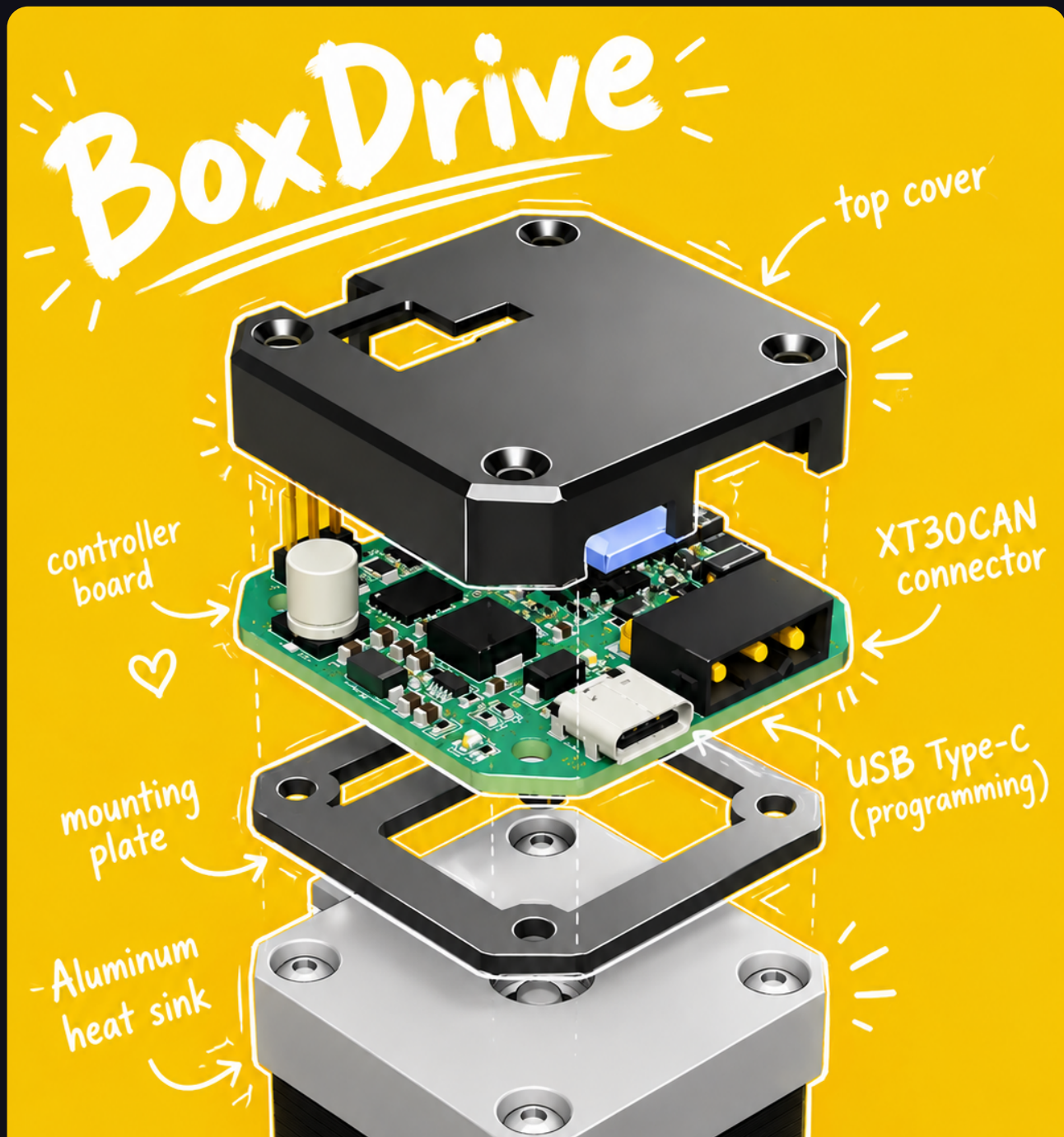
USB-C

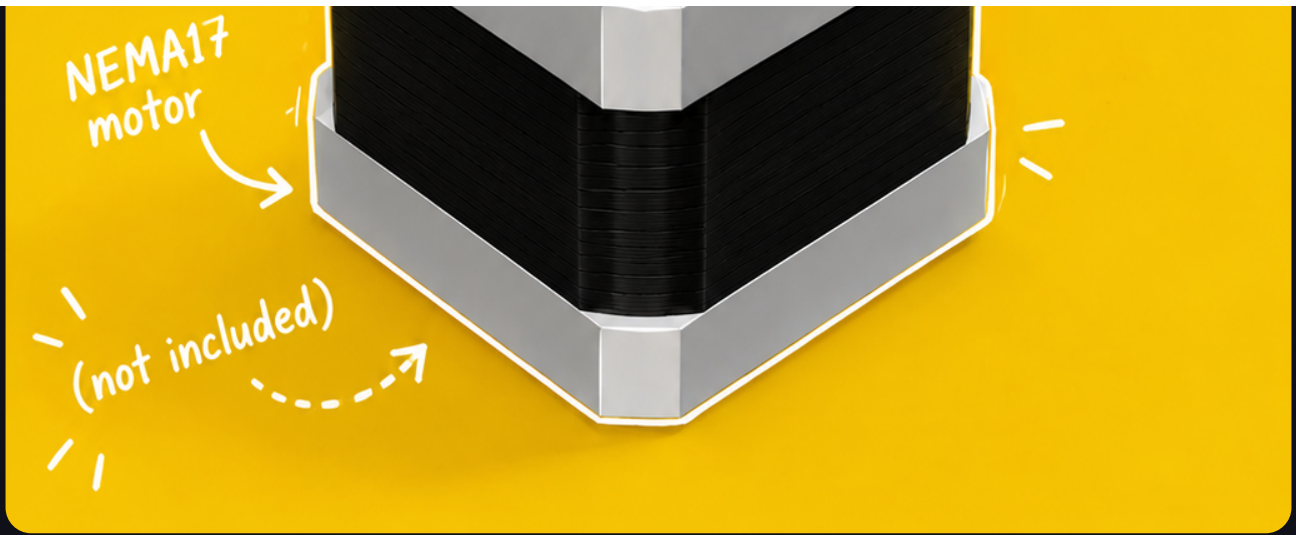
Native USB programming and debug – Arduino, PlatformIO, ESP-IDF.

NEMA 17 MOUNT

$42.31 \times 42.31 \times 9.1$ mm with 4× M3 holes on the 31 mm pattern.

THIS IS WHAT YOU GET.





- ★ BoxDrive
- ★ XT30CAN cable
- ★ Aluminum heat sink
- ★ Casing
- ★ 4 M3 Screws
- ★ Magnet



Manual

GET YOUR BOXDRIVE

BUY

In stock now. Ships in 2-3 business days.



BOXPOX

DATASHEET

SHOP

CART

Address: D-180C, Phase 8B, Industrial Area, Sector 74,
Sahibzada Ajit Singh Nagar, Punjab 160055

Phone: 9041579564, 7888601710

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